

Verification as a Costed Action: When Checking Beats Containing Under Attack–Fault Ambiguity

Chiheb Mejrissi

Abstract

When an anomaly on a trading gateway may come from a compromised account or from a benign fault, an operator who sees only order flow can be unable to tell the two apart. Verification—validating the credential or session, or calling the client—reads a different channel, authorization, and is therefore not bound by what order flow can reveal. We model verification as a costed action with sensitivity, specificity, a fixed cost and a delay during which a true compromise keeps doing harm. Transposing the threshold approach of clinical decision analysis, we show that verification pays exactly on an interval of beliefs around the passive containment threshold, which yields a leave / verify / contain rule with two thresholds in closed form. Because harm during the delay accrues only under compromise, the cost of checking grows with the belief, and the interval shrinks to nothing at a critical delay that we give in closed form: an operational time budget beyond which no check should be run. The interval is a set of beliefs and does not depend on the prior; what depends on the prior is how much of a population’s posterior mass falls inside it, and the saving from verification vanishes as the prior of compromise goes to zero. An exploratory empirical study on the classifier posteriors of a companion order-flow study examines where targeted verification beats the best constant policy in the cells where passive classification adds nothing.

Keywords: costly verification, value of information, threshold decision rule, intrusion response, attack–fault ambiguity, electronic trading

1. Introduction

An operator of an electronic trading gateway sees an account submit an unusual pattern of orders. The pattern may come from an adversary who

has taken over the account or the session, or from a benign malfunction of the client’s own systems. The operator can *contain*—suspend the session, pull the orders—or *leave* trading running. Containing a fault disrupts a legitimate client for nothing; leaving a compromise lets the adversary keep trading. When the two causes produce order flow that looks the same, no amount of order-flow analysis settles the matter.

The separate channel. A companion study, Paper 1 [1], makes this precise. In its linear-Gaussian order-flow model, when the confounding condition holds, order-flow observation cannot separate a compromise from a fault; the result of Paper 1 used here is its Proposition 1, by which an operator who sees only order flow can then do no better than act on the prior, and pays a cost of ambiguity, over an operator who knows the cause, equal to the expected regret of the better of the two terminal actions taken at the prior (restated for our loss table as Corollary 1(i)). That limit constrains only decision rules that are functions of the order flow. Verification reads another channel, *authorization*, and under a conditional-independence assumption it composes with whatever the passive analysis concludes (Lemma 2); Section 3.4 develops the argument. This is the principled reason a costly check can beat every passive classifier: it is not subject to the bound that limits them.

When a check pays. A check is not free. It costs analyst time and client friction, and it takes time during which a real adversary keeps trading. We model verification as an action with sensitivity s , specificity t , a fixed cost c_v and a delay τ during which a true compromise causes harm at rate h_K , on top of a general loss table (C_K, C_F, H_K, H_F) for the two terminal actions. The passive operator contains when the belief in a compromise exceeds a threshold q^* (Lemma 1). The value of a check is a tent centred on q^* (Lemma 3), and the check pays exactly where the tent exceeds its cost. The result is a leave / verify / contain rule with two thresholds $q_1 < q^* < q_2$ in closed form (Theorem 1). This rule is the threshold approach of Pauker and Kassirer [2] transposed to attack-fault triage, and we claim neither it nor its value-of-information step as new (Section 2).

What the setting adds is the shape of the cost. Because harm during the delay accrues only if the cause is a compromise, the expected cost of checking, $c_v + q h_K \tau$, grows with the belief. As the delay grows, the interval (q_1, q_2) shrinks and vanishes at a critical delay τ^* (Theorem 2). This too follows directly from the tent, and belief-dependent costs of delay have precedents

(Section 2). Its use here is operational: τ^* is a time budget for a verification procedure. A check that takes longer than τ^* should never be run, at any belief.

What the prior changes, and what it does not. The verification interval is a set of beliefs; it does not depend on the prior probability of compromise. What depends on the prior is how much of the posterior mass of a population of cases falls inside it, and hence how much targeted verification saves on that population. We define these per-population quantities and show that the saving vanishes as the prior goes to zero, and so does the mass in the interval unless the check is both free and never flags a fault (Lemma 4). A low prior therefore does not by itself make verification more valuable; whether a check helps in a given population is an empirical question about where that population’s posteriors fall relative to q_1 .

Contributions.

1. *The separate-channel argument.* Verification that reads authorization is conditionally independent of order flow given the cause, so it composes with any passive posterior (Lemma 2) and lowers the confounded cost of ambiguity exactly when the prior lies in the verification interval (Corollary 1).
2. *The critical delay.* A closed-form τ^* at which the verification interval vanishes, with nested intervals below it and the manner of vanishing characterised (Theorem 2). It is an operational corollary of the value-of-information tent under a cost of delay borne only by a true compromise.
3. *An exploratory test on the failure cells of Paper 1.* In the cells where Paper 1’s classifier does no better than the best constant policy, we examine whether targeted verification beats that policy, and over which region of (s, t, c_v, τ) , against the region predicted from Theorem 1 and the cells’ posteriors. The test was specified after Paper 1’s cost table for these cells had been examined, so it is reported as exploratory, not as a pre-registered test.

Section 2 positions the paper. Section 3 sets out the model and the separate-channel assumption, and Section 4 derives the rule, the critical delay and the per-population quantities. Section 5 describes the empirical study. Section 6 discusses scope and limitations, and Section 7 concludes.

2. Related work

Threshold analysis and the value of information. The leave / verify / contain rule of Section 4 is the threshold approach of Pauker and Kassirer [2] transposed to attack–fault triage. There, a clinician’s probability of disease is split by a *testing threshold* and a *test-treatment threshold* into three regions: withhold treatment, test and follow the result, treat without testing. The thresholds follow from the test’s sensitivity and specificity, its cost or risk, and the benefit and harm of treatment. Our q_1 and q_2 play the same roles, with contain in place of treat and a general loss table (C_K, C_F, H_K, H_F) in place of the clinical utilities. The underlying calculation—a single experiment is worth running exactly when its expected value of sample information exceeds its cost, and for two actions and two states that value is piecewise linear and vanishes wherever no outcome can change the action—is classical Bayesian decision theory [3]. We therefore claim neither the rule nor the tent shape of the value of information (Lemma 3) as new.

Costs of delay that depend on the state. The feature that distinguishes our setting is a cost of checking that falls only on one cause: an adversary keeps trading while the check runs, a fault does not get worse. Costs of delay that depend on the belief about the underlying state are a central object of time-critical decision analysis, where the expected cost of deferring an action is computed from a distribution over states and a time-dependent utility and weighed against the value of gathering information [4, 5]. In medical decision making, Schroeder et al. [6] use the threshold framework to trade the delay of a more accurate send-out test against the clinical benefit lost while waiting for the result; this is the closest precedent we know for the delay term $q h_K \tau$ of our model. At the population level, the value of further research is likewise eroded by the decisions made while it is pending [7]. Given the tent, the critical delay τ^* of Theorem 2 follows in one step, and we present it as an operational corollary for attack–fault triage rather than as a new decision-theoretic result.

Costly and active information acquisition. Choosing which observations to buy before acting is studied as Bayesian active learning and adaptive diagnosis, where tests are selected sequentially to identify a hypothesis at least cost [8], and as optimal stopping with costly sequential information, where the reversibility of the terminal decision shapes when to stop [9]. Our problem

is the one-shot special case with two causes, two terminal actions and a single available check. We keep it one-shot on purpose: it gives the thresholds and the critical delay in closed form, and the sequential version is a separate problem.

Intrusion response and alert verification. Intrusion response has long been cast as a cost-sensitive decision, in which the damage an attack would cause is weighed against the cost of responding [10], and later as a game or a partially observed control problem [11, 12, 13, 14]. Alert verification checks whether an intrusion attempt actually succeeded before an alert is raised [15]; it is a technical filter, with no cost, delay or threshold attached. Separating cyberattacks from physical faults has been studied from the observation side, asking when the two are distinguishable from sensor data [16, 17]. Our contribution sits between these lines: a verify-then-respond decision layer with an explicit accuracy, cost and cause-borne delay, justified by the fact that the check reads a channel the observation-side analysis does not cover.

Prior shift and cost-sensitive classification. The empirical study of Section 5 uses classifier scores as passive posteriors. Scores trained at one class balance are moved to another prior by the standard correction [18], and a calibrated posterior is turned into a cost-minimising decision by thresholding it at the cost ratio [19]; the passive threshold q^* of Lemma 1 is that threshold for our loss table.

3. Model

An operator watches an anomaly on a trading gateway and must decide what caused it. Two causes are possible: a *compromise* K , in which an adversary is acting through a legitimate account or session, and a *fault* F , in which the same symptoms come from a benign malfunction. The operator can *contain* (suspend the account or session, pull the orders) or *leave* (let trading continue). Before committing to either, the operator may also *verify*: query a source of evidence that the order flow does not carry. This section sets out the decision problem; Section 4 solves it.

3.1. Loss table and passive risk

The operator’s belief that the cause is a compromise is $q \in [0, 1]$. Without verification, the operator picks the terminal action with the lower expected loss.

Definition 1 (Loss table, passive risk). There are two causes, compromise K and fault F , and two terminal actions, *contain* and *leave*. Containing costs C_K under K and C_F under F ; leaving costs H_K under K and H_F under F . For a belief $q = P(K) \in [0, 1]$ the *passive risk* is

$$R(q) = \min\{q C_K + (1 - q) C_F, q H_K + (1 - q) H_F\}.$$

Write $a = H_K - C_K$ and $b = C_F - H_F$.

The four entries of the loss table are kept general. Containing a compromise (C_K) costs the disruption of a legitimate client plus the operational cost of the response; containing a fault (C_F) costs the same disruption with nothing gained; leaving a compromise (H_K) costs the harm the adversary goes on to do; leaving a fault (H_F) costs whatever the malfunction costs if it is not interrupted. The quantity a is what containment saves when the cause is a compromise, and b is what it wastes when the cause is a fault. The cost matrix of Paper 1 [1] is the special case $H_F = 0$, with Paper 1's containment cost, disruption and harm in the roles of C_K , C_F and H_K .

Nothing in the model depends on the prior probability of compromise except through the belief q . The prior enters when cases are drawn with a given prior: for a single case whose order flow is uninformative, in Section 4.6 (Definition 4 and Corollary 1, where the belief is the prior), and for a population of cases, in Section 4.8.

3.2. Verification

Verification is a costed action with an uncertain outcome. On a trading gateway it is, for example, a validation of the credential or session that submitted the orders, or a call from the desk to the client to confirm the activity.

Definition 2 (Verification action). Verification returns an outcome $Z \in \{+, -\}$ with sensitivity $s = P(Z = + \mid K)$ and specificity $t = P(Z = - \mid F)$. It costs c_v and takes a delay τ ; during the delay a true compromise causes harm at rate h_K . From belief q ,

$$P(+)=qs+(1-q)(1-t), \quad P(-)=q(1-s)+(1-q)t,$$

$$q^+ = \frac{qs}{P(+)}, \quad q^- = \frac{q(1-s)}{P(-)}$$

(a posterior whose outcome has probability zero is never used; see Definition 3).

The sensitivity s is the probability that verification flags a compromise that is present; the specificity t is the probability that it clears a fault. The cost c_v is paid whatever the cause: analyst time, the call, the friction imposed on the client. The delay τ is the time the check takes, and the harm rate h_K is what an adversary who is really present gains per unit of that time. A fault does not grow worse because the operator is checking; a compromise does.

Definition 3 (Verify-first risk, value of information, verification region). The *verify-first risk* is

$$R_v(q) = c_v + q h_K \tau + P(+) R(q^+) + P(-) R(q^-),$$

where $P(z) R(q^z)$ is read as $\min\{P(z, K) C_K + P(z, F) C_F, P(z, K) H_K + P(z, F) H_F\}$ with $P(z, K) = P(z \mid K) q$ and $P(z, F) = P(z \mid F) (1 - q)$ (equal to the product when $P(z) > 0$, and zero when $P(z) = 0$). The *value of information* is $V(q) = R(q) - [P(+) R(q^+) + P(-) R(q^-)]$, the *verification cost* is $\kappa(q) = c_v + q h_K \tau$, and the *verification region* is

$$S = \{q \in [0, 1] : R_v(q) < R(q)\} = \{q \in [0, 1] : V(q) > \kappa(q)\}.$$

Ties are resolved in favour of the passive action: the operator verifies only when verifying is strictly cheaper.

The verify-first risk adds three terms to the expected terminal loss after the update: the cost of the check, the harm expected during the delay, and the expected passive risk at the updated belief. The value of information V is the familiar reduction in terminal risk that the outcome buys [3]; the verification cost κ collects what the check itself costs. Because harm during the delay accrues only under K , κ grows with the belief: checking is more expensive precisely when a compromise is more likely. The verification region S is the set of beliefs at which the check pays for itself.

3.3. Assumptions

Assumption 1 (Cost orderings). $H_K > C_K$ and $C_F > H_F$; that is, $a > 0$ and $b > 0$: leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it.

This excludes the uninteresting cases in which one terminal action dominates the other whatever the cause. On a gateway it says that an unchecked adversary does more damage than a response costs, and that suspending

a healthy client costs more than letting a benign fault run. Remark 1 in Section 4 records what happens when it fails.

Assumption 2 (Separate channel). Verification reads authorization (credential or session validity), not order flow. Its outcome Z is conditionally independent of the order-flow observation Y given the cause: $P(Z, Y | c) = P(Z | c) P(Y | c)$ for $c \in \{K, F\}$, and (s, t) do not depend on Y .

Assumption 2 is the conceptual core of the paper and is discussed in Section 3.4.

Assumption 3 (Informative one-shot verification). $s, t \in [0, 1]$ with $s + t > 1$; $c_v \geq 0$, $\tau \geq 0$, $h_K \geq 0$. Verification is available once; after its outcome the operator takes a terminal action (contain or leave) and cannot verify again.

The condition $s + t > 1$ says that a positive outcome is more likely under a compromise than under a fault; a check that fails it carries no information (Remark 1). The one-shot restriction matches a triage step in which one check is run and the operator then acts. Repeated or sequential checking is a stopping problem and is outside the scope of this paper.

Assumption 4 (Delay accounting). The harm $h_K \tau$ accrued by a true compromise during the verification delay is *additional* to the terminal cost (C_K or H_K) incurred afterwards, whichever terminal action is taken; the terminal costs C_K, C_F, H_K, H_F and the verification characteristics s, t do not depend on τ ; a fault accrues no additional cost during the delay; and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 4 fixes how the delay is charged. The harm accrued during the check is paid in addition to the terminal cost, whatever the operator then does; this is the conservative accounting for verification, since it charges the delay in full even when the terminal cost H_K might already include part of it. The check does not change the loss table or its own accuracy, a fault costs nothing extra while it is checked, and the operator does not update on further order flow while waiting for the outcome.

3.4. Why verification is not bound by the order-flow limit

In its linear-Gaussian order-flow model, when the confounding condition holds, Paper 1 [1] shows that order-flow observation cannot separate a compromise from a fault, and that a passive operator who sees only order flow then pays a cost of ambiguity that no classifier can remove. That limit is a statement about the law of the order-flow observation Y under the two

causes; it constrains every decision rule that is a function of Y alone. It does not constrain evidence that is not a function of Y .

Verification is such evidence. A credential or session check reads the authorization channel: whether the entity submitting the orders is the one entitled to submit them. Assumption 2 states that, given the cause, the outcome of that check is independent of what the order flow shows, and that its accuracy (s, t) does not depend on the order flow. Operationally, whether a session token is valid does not depend on how the orders it submitted look. Under this assumption verification composes with whatever the passive analysis concludes: Lemma 2 in Section 4 shows that the joint posterior is the verification update applied to the order-flow posterior, with the same characteristics (s, t) whatever the order-flow law. In particular it applies when the order-flow laws under K and F coincide. That is the case in which Paper 1’s passive operator can do no better than act on the prior; Corollary 1 in Section 4 states when verification brings the cost of ambiguity strictly below Paper 1’s value there.

Assumption 2 is a modelling choice, not a consequence of the setting. Without it the conditional law of the check’s outcome given the cause may depend on the order-flow observation, the verification update is no longer a function of the order-flow posterior alone, and Lemma 2 no longer holds.

4. The two-threshold rule and the critical delay

This section derives when verification pays. The passive rule has a single threshold (Lemma 1). Verification composes with any order-flow posterior (Lemma 2), and its value is a tent centred on that threshold (Lemma 3). Comparing the tent with the cost of the check gives a leave / verify / contain rule with two thresholds (Theorem 1), and a critical delay beyond which the check never pays (Theorem 2). Corollary 1 applies these results to the confounded case, and Remark 1 records the degenerate parameters. Lemma 4 and Proposition 1 then turn from a single belief to a population of cases with a given prior, which is where the prior enters. Proofs are collected in Section 4.9.

4.1. The passive threshold

Lemma 1 (Passive threshold). Under Assumption 1, let $q^* = b/(a + b) \in (0, 1)$. Leaving is strictly optimal for $q < q^*$, containing is strictly

optimal for $q > q^*$, and

$$R(q) = q H_K + (1 - q) H_F - (a + b) (q - q^*)^+,$$

so R is continuous, concave and piecewise linear on $[0, 1]$ with a single kink at q^* .

The operator who cannot verify contains exactly when the belief in a compromise exceeds q^* : the share of the total stakes $a + b$ that a wrong containment wastes. A response that is costly to a healthy client (large b) pushes q^* up; an adversary who does much damage if left alone (large a) pulls it down. At q^* the two terminal actions are equally good, and the passive risk has its only kink.

4.2. Verification composes with the passive posterior

Lemma 2 (Verification composes with any passive posterior). Under Assumption 2, let $q(y) = P(K \mid Y = y)$ be the passive (order-flow) posterior. Then $P(K \mid Y = y, Z = z)$ equals the verification update of Definition 2 applied to $q = q(y)$: it is $q(y)^+$ if $z = +$ and $q(y)^-$ if $z = -$. In particular the characteristics (s, t) , and hence V , S and all results below, are the same whatever the order-flow law, including when the order-flow laws under K and F coincide.

Lemma 2 separates the two stages of the decision. The operator first reduces the order-flow evidence to a belief $q(y)$, with whatever classifier or model is available, and then applies the verification analysis below to that belief, with the same (s, t) . The verification rule need not be re-derived for each classifier, and it remains meaningful when the order flow carries no information about the cause (Section 3.4). All results below are stated for a belief q and hold at $q = q(y)$.

4.3. The value of information is a tent

Lemma 3 (The value of information is a tent). Under Assumptions 1 and 3,

$$V(q) = \max\{0, \min\{G^+(q), G^-(q)\}\},$$

where

$$G^+(q) = q s a - (1 - q)(1 - t) b, \quad G^-(q) = (1 - q) t b - q(1 - s) a.$$

Consequently, with

$$q_L = \frac{(1-t)b}{sa + (1-t)b}, \quad q_U = \frac{tb}{tb + (1-s)a},$$

(i) $0 \leq q_L < q^* < q_U \leq 1$; (ii) $V(q) > 0$ if and only if $q \in (q_L, q_U)$, equivalently if and only if $q^- < q^* < q^+$ (the two posteriors fall strictly on opposite sides of q^*); (iii) V is affine and strictly increasing on $[q_L, q^*]$, affine and strictly decreasing on $[q^*, q_U]$, and zero outside (q_L, q_U) ; (iv) V attains its maximum uniquely at q^* , with

$$V(q^*) = (s+t-1) \frac{ab}{a+b} = (s+t-1) a q^* > 0.$$

Lemma 3 is the standard value-of-information calculation for two actions and two states [3]. A check is worth something only when its outcome can change the terminal action: a positive outcome must move the belief above q^* and a negative outcome below it. Far from the threshold neither outcome can overturn the passive decision, and the check is worthless however accurate it is. The value rises linearly from q_L to its peak at q^* and falls linearly to q_U . The height of the peak is the product of the Youden index $s+t-1$ and the quantity $a q^* = b(1-q^*)$, which is the expected regret at the threshold, relative to an operator who knows the cause, of either terminal action.

4.4. The two-threshold rule

Theorem 1 (Two-threshold leave / verify / contain rule). Under Assumptions 1, 3 and 4, the verification region S is an open interval (possibly empty). It is non-empty if and only if

$$V(q^*) > c_v + q^* h_K \tau,$$

and then $S = (q_1, q_2)$ with $q_L \leq q_1 < q^* < q_2 \leq q_U$, where

$$q_1 = \frac{c_v + (1-t)b}{sa + (1-t)b - h_K \tau}, \quad q_2 = \frac{tb - c_v}{tb + (1-s)a + h_K \tau}$$

(both denominators are positive when $S \neq \emptyset$). The optimal one-shot policy is: *leave* if $q \leq q_1$, *verify* if $q_1 < q < q_2$ and then contain on $Z = +$ and leave on $Z = -$, *contain* if $q \geq q_2$ (with $q_1 = q_2 = q^*$ and the passive rule of Lemma 1 when $S = \emptyset$). When $c_v = 0$ and $\tau h_K = 0$, $S = (q_L, q_U)$.

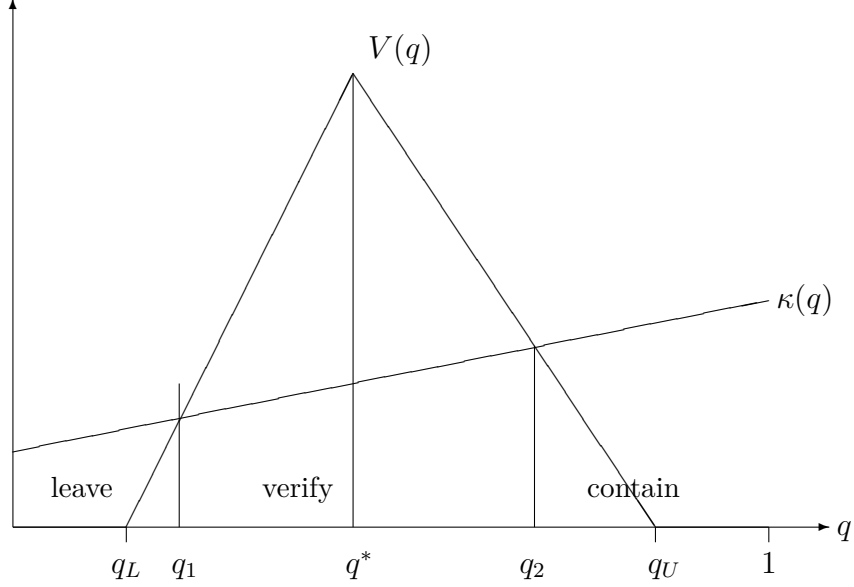


Figure 1: Schematic of Lemma 3 and Theorem 1 (not to scale; no parameter values). The value of information V is a tent on (q_L, q_U) with its peak at the passive threshold q^* ; the verification cost κ increases with the belief because harm during the delay accrues only under compromise. Verification pays on (q_1, q_2) , where the tent lies above the cost line. Raising the delay τ tilts the cost line upward and shrinks the interval (Theorem 2).

Theorem 1 is the threshold rule of Pauker and Kassirer [2], with q_1 and q_2 in the roles of the testing and test-treatment thresholds (Section 2); what the transposition adds is the cost of the check that grows with the belief, through the delay term $q h_K \tau$.

For an operator the rule reads as follows. Below q_1 the check costs more than it is expected to save, and the operator leaves. Above q_2 a compromise is likely enough that the operator contains without checking. In between, the operator verifies and follows the outcome. The verification interval always contains the passive threshold q^* when it is non-empty, and it lies inside the interval (q_L, q_U) on which the check has any value at all. Its two ends move inward as the check becomes more expensive or slower: c_v raises the lower end and lowers the upper one, and the delay term $h_K \tau$ enters both denominators.

4.5. The critical delay

Theorem 2 (Critical delay). Under Assumptions 1, 3 and 4, write $S(\tau)$ for the verification region at delay τ , all other parameters fixed. (i) *Nesting.* $\tau < \tau'$ implies $S(\tau') \subseteq S(\tau)$. If $h_K > 0$ and $V(q^*) > c_v$, then $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$ (part (ii)), and on $[0, \tau^*)$ the end point $q_2(\tau)$ of Theorem 1 is strictly decreasing in τ , and $q_1(\tau)$ is strictly increasing in τ unless $c_v = 0$ and $t = 1$, in which case $q_1(\tau) = 0$ on $[0, \tau^*)$. For $\tau \geq \tau^*$, $S(\tau) = \emptyset$ and the convention $q_1 = q_2 = q^*$ of Theorem 1 applies. (ii) *Critical delay.* If $h_K > 0$ and $V(q^*) > c_v$, let

$$\tau^* = \frac{V(q^*) - c_v}{q^* h_K} = \frac{(s + t - 1)a - c_v/q^*}{h_K} > 0.$$

Then $S(\tau) \neq \emptyset$ if and only if $\tau < \tau^*$. (iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for every $\tau \geq 0$. If $h_K = 0$, $S(\tau)$ does not depend on τ . (iv) *How the interval vanishes.* If $c_v > 0$ or $t < 1$, then $q_1 \rightarrow q^*$ and $q_2 \rightarrow q^*$ as $\tau \uparrow \tau^*$: the interval shrinks continuously to the point q^* . If $c_v = 0$ and $t = 1$, then $S(\tau) = (0, q_2(\tau))$ for $\tau < \tau^*$ with $q_2(\tau) \downarrow q^*$, and $S(\tau^*) = \emptyset$: the interval vanishes discontinuously.

Theorem 2 is an operational corollary of Lemma 3 and Theorem 1. Once the value of information is known to be a tent peaking at q^* , and the cost of the check is affine and increasing in the belief, the check pays at some belief if and only if it pays at q^* ; solving that inequality for the delay gives τ^* . Its precedents are discussed in Section 2; what Theorem 2 packages for attack–fault triage is the critical delay as a single closed-form quantity.

For an operator, τ^* is a time budget for the check. A verification procedure that takes longer than τ^* should never be run, at any belief: the operator should act on the passive rule instead. Below τ^* the verification interval shrinks as the check slows down, and the operator verifies over a narrower band of beliefs around q^* . The budget grows with the accuracy of the check (through $s + t - 1$) and shrinks with the harm rate h_K and with the fixed cost c_v . If the fixed cost alone exceeds the peak value of information, no delay is short enough. In the limiting case of a free check that never flags a fault ($c_v = 0$ and $t = 1$), the interval reaches down to zero belief and disappears all at once at τ^* ; otherwise it closes continuously onto q^* .

4.6. Comparison with the passive cost of ambiguity

The results above are stated for a belief. To compare verification with the passive analysis of Paper 1 [1], the decision problem is stated over policies

that act on the order-flow observation.

Definition 4 (Decision problem with verification; verify-first cost of ambiguity). The cause M is K with prior probability $p = P(K) \in (0, 1)$ and F otherwise. Given M , the order-flow observation Y has law P_M^Y , and the verification outcome Z has the law of Definition 2, jointly as in Assumption 2. Write $L(\text{contain}, K) = C_K$, $L(\text{contain}, F) = C_F$, $L(\text{leave}, K) = H_K$, $L(\text{leave}, F) = H_F$ (Definition 1). A *verify-first policy* is a quadruple of measurable maps $\delta, \pi_0, \pi_+, \pi_-$ from observations y to $[0, 1]$: with probability $\delta(y)$ the operator verifies and then contains with probability $\pi_Z(y)$; otherwise it contains with probability $\pi_0(y)$. Its expected cost is

$$\begin{aligned} J(\delta, \pi) = \mathbb{E} \bigg[& (1 - \delta(Y)) \left(\pi_0(Y) L(\text{contain}, M) \right. \\ & \left. + (1 - \pi_0(Y)) L(\text{leave}, M) \right) \\ & + \delta(Y) \left(c_v + \mathbb{1}[M = K] h_K \tau + \pi_Z(Y) L(\text{contain}, M) \right. \\ & \left. + (1 - \pi_Z(Y)) L(\text{leave}, M) \right) \bigg]. \end{aligned}$$

A *passive policy* is a verify-first policy with $\delta \equiv 0$. Let R_Y^* and $R_{Y,v}^*$ be the infima of J over passive and over verify-first policies, $R_{\text{oracle}} = p \min\{C_K, H_K\} + (1 - p) \min\{C_F, H_F\}$ the expected cost of an operator who observes M , and

$$\text{CoA}^* = R_Y^* - R_{\text{oracle}}, \quad \text{CoA}_v^* = R_{Y,v}^* - R_{\text{oracle}}$$

the passive and the verify-first cost of ambiguity.

Corollary 1 (Verification lowers the confounded cost of ambiguity). Under Assumptions 1–4, in the setting of Definition 4, suppose the order-flow laws coincide, $P_K^Y = P_F^Y$ (the confounded case). Then: (i) $\text{CoA}^* = \min\{p a, (1 - p) b\} > 0$; (ii) the verify-first cost of ambiguity is

$$\text{CoA}_v^* = \min\{ \text{CoA}^*, E(p) \},$$

$$E(p) = c_v + p h_K \tau + p(1 - s) a + (1 - p)(1 - t) b,$$

equivalently $\text{CoA}_v^* = \text{CoA}^* - (V(p) - \kappa(p))^+$, and it is attained by the policy of Theorem 1 applied at the belief $q = p$, whatever Y ; (iii) $\text{CoA}_v^* < \text{CoA}^*$ if and only if $p \in S$, that is, if and only if $V(q^*) > c_v + q^* h_K \tau$ and $q_1 < p < q_2$ (Theorem 1); if $h_K > 0$ and $V(q^*) > c_v$, this holds if and only if $\tau < \tau^*$ and $q_1(\tau) < p < q_2(\tau)$ (Theorem 2), and if $V(q^*) \leq c_v$ it holds for no p and no τ ;

(iv) over $p \in (0, 1)$, the reduction $\text{CoA}^* - \text{CoA}_v^*$ is largest at $p = q^*$, where it equals $(V(q^*) - c_v - q^* h_K \tau)^+$; (v) $\text{CoA}_v^* = 0$ if and only if $s = t = 1$, $c_v = 0$ and $h_K \tau = 0$. Under the correspondence of Paper 1, Definition D8 ($H_F = 0$), CoA^* in (i) is the cost of ambiguity of Paper 1, Proposition 1, so (iii) states when verification brings the cost of ambiguity strictly below that value.

In the confounded case the order flow carries no information about the cause, and the passive cost of ambiguity in part (i), stated for the general loss table, reduces to the value of Paper 1's Proposition 1 when $H_F = 0$. Corollary 1 states when verification brings it strictly lower: exactly when the prior lies in the verification region of Theorem 1, so, for a harmful compromise and a check whose fixed cost is below the peak value of information, only when the check is faster than τ^* . The reduction is largest when the prior sits at the passive threshold, and the cost of ambiguity is removed entirely only by a perfect and free check that accrues no harm during its delay. The optimal verify-first policy ignores the order flow, which here carries no information, and applies the rule of Theorem 1 to the prior.

4.7. Degenerate parameters

Remark 1 (Degenerate parameters). (a) If $s+t = 1$, then $q^+ = q^- = q$ whenever defined, $V \equiv 0$, and $S = \emptyset$ for all $c_v \geq 0$, $\tau \geq 0$. (b) Assumption 1 fails exactly when $a \leq 0$ or $b \leq 0$, that is, in one of two cases. (b1) *Weak signs mixed or zero:* $a \leq 0 \leq b$ or $b \leq 0 \leq a$ (so $q^* = b/(a+b) \notin (0, 1)$ or is undefined). Then one terminal action is weakly optimal at every belief, R is affine, $V \equiv 0$, and $S = \emptyset$. (b2) *Both negative:* $a < 0$ and $b < 0$. Then $q^* = b/(a+b) \in (0, 1)$, containing is strictly optimal for $q < q^*$ and leaving for $q > q^*$. Exchanging the names of the two terminal actions gives a problem with the same R , V , κ and S that satisfies Assumption 1 with (a, b) replaced by $(-a, -b)$ and the same q^* ; every result below then holds for that problem, i.e. for the original one with (a, b) replaced by $(-a, -b)$ and *contain* and *leave* exchanged (in Theorem 1: contain if $q \leq q_1$, verify if $q_1 < q < q_2$ and then leave on $Z = +$ and contain on $Z = -$, leave if $q \geq q_2$). (c) The requirement $c_v \geq 0$ in Assumption 3 is needed for Theorem 1: if $c_v < 0$, S can contain beliefs near 0 where $V = 0$ and can then be disconnected.

Parts (a) and (b) say that the results above need a check that is informative and a decision that is not settled in advance. Part (c) explains why Assumption 3 requires a non-negative cost: a check that pays the operator to run it would be worth running at beliefs where its outcome cannot change the action, and the verification region could then fail to be an interval.

4.8. Populations of cases: what the prior changes

The verification region $S = (q_1, q_2)$ is a set of beliefs. Its end points depend on the loss table and on (s, t, c_v, τ, h_K) , and not on the prior probability of compromise. An operator facing a population of cases—for instance all the anomalies raised under one market design and one base rate of compromise—sees instead a distribution of passive posteriors, and the prior moves that distribution. The following quantities measure how much of it falls in the verification region and what verifying there saves.

Definition 5 (Verification mass, saving and recovered share of a cell). An *order-flow model* c is a pair of order-flow laws P_K^c, P_F^c on a measurable space, with densities f_K^c, f_F^c with respect to a σ -finite measure ν dominating both (e.g. $\nu = P_K^c + P_F^c$). A *cell* (c, p) is an order-flow model c together with a prior $p \in (0, 1)$; in Paper 1's grid, c is a market design in one regime and p is a prior of the grid. In the setting of Definition 4 with $P_K^Y = P_K^c, P_F^Y = P_F^c$ and prior $p \in (0, 1)$, write $P_p = p P_K^c + (1 - p) P_F^c$ for the law of Y and

$$q_p(y) = \frac{p f_K^c(y)}{p f_K^c(y) + (1 - p) f_F^c(y)}$$

for the passive posterior (defined where the denominator is positive, a set of full P_p -measure). The *verification mass*, the *saving* and the *recovered share* of the cell (c, p) are

$$m_c(p) = P_p(q_p(Y) \in S), \quad D_c(p) = \mathbb{E}_p[(V - \kappa)^+(q_p(Y))],$$

$$\rho_c(p) = \frac{D_c(p)}{\text{CoA}_c^*(p)},$$

with S, V, κ as in Definition 3 and $\text{CoA}_c^*(p)$ the passive cost of ambiguity of Definition 4 for this cell; $\rho_c(p)$ is defined when $\text{CoA}_c^*(p) > 0$. Where $f_K^c, f_F^c > 0$, $\text{logit } q_p(y) = \text{logit } p + \Lambda_c(y)$ with $\Lambda_c = \log(f_K^c/f_F^c)$.

Lemma 4 (Saving identity and low-prior limit). Under Assumptions 1–4, in the setting of Definition 5: (i) *Saving.* $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p) = R_Y^* - R_{Y,v}^*$, where $\text{CoA}_{v,c}^*(p)$ is the verify-first cost of ambiguity of Definition 4 for this cell; it is attained by the policy that verifies iff $q_p(Y) \in S$, then contains on $Z = +$ and leaves on $Z = -$, and otherwise takes the passive action of Lemma 1 at $q_p(Y)$. Moreover $0 \leq D_c(p) \leq \min\{\text{CoA}_c^*(p), V(q^*) m_c(p)\}$, so $0 \leq \rho_c(p) \leq 1$ where defined. (ii) *The saving vanishes at low priors.* For every order-flow model c , $D_c(p) \rightarrow 0$ as $p \rightarrow 0$.

(iii) *The mass vanishes at low priors when $q_1 > 0$.* If $S = \emptyset$, then $m_c \equiv 0$. If $S = (q_1, q_2) \neq \emptyset$ and $q_1 > 0$ (equivalently, $c_v > 0$ or $t < 1$), then $m_c(p) \rightarrow 0$ as $p \rightarrow 0$. (iv) *The exception.* If $S \neq \emptyset$, $c_v = 0$ and $t = 1$ (so $q_1 = 0$), then $m_c(p) \rightarrow P_F^c(f_K^c(Y) > 0)$ as $p \rightarrow 0$. No condition on the order-flow model is needed beyond Definition 5: the densities are finite, and $f_F^c > 0$ holds P_F^c -almost surely by construction.

Part (i) identifies the saving of targeted verification with the reduction in the cost of ambiguity, and bounds it by the peak value of information times the mass in the interval. Parts (ii) and (iii) state what happens when compromise becomes rare. As the prior falls, the passive posteriors move towards zero and, in the limit, below the lower threshold q_1 whenever the check has a fixed cost or can flag a fault, so the mass in the interval and the saving both vanish. A low prior therefore does not make verification more valuable on its own: what matters is whether the upper tail of the posteriors still reaches above q_1 . Part (iv) is the only exception for the mass: for a free check that never flags a fault the interval reaches down to zero belief (Theorem 2(iv)) and the mass has a positive limit, although the saving still vanishes by part (ii).

Lemma 4 is the reason the empirical question of Section 5 is posed per population. Whether targeted verification saves anything in a given population depends on where that population's posteriors fall relative to q_1 , which the theory does not decide. For a Gaussian observation model the quantities of Definition 5 have closed forms.

Proposition 1 (Closed forms for a Gaussian order-flow model).

Under Assumptions 1–4, let the order-flow model c of Definition 5 be Gaussian: $Y \mid M \sim \mathcal{N}(\mu_M, \Sigma_Y)$, $M \in \{K, F\}$, with common covariance $\Sigma_Y \succ 0$ and whitened separation $d = \sqrt{(\mu_K - \mu_F)^\top \Sigma_Y^{-1} (\mu_K - \mu_F)} > 0$. Suppose $S = (q_1, q_2) \neq \emptyset$. Write $\ell = \text{logit } p$, $\ell_1 = \text{logit } q_1$, $\ell^* = \text{logit } q^* = \log(b/a)$, $\ell_2 = \text{logit } q_2$ (with $\text{logit } 0 = -\infty$, $\text{logit } 1 = +\infty$, $\Phi(\pm\infty) = 1, 0$ accordingly), and for $-\infty \leq u < v \leq +\infty$

$$\begin{aligned}\Pi_K(u, v) &= \Phi\left(\frac{v - \ell}{d} - \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} - \frac{d}{2}\right), \\ \Pi_F(u, v) &= \Phi\left(\frac{v - \ell}{d} + \frac{d}{2}\right) - \Phi\left(\frac{u - \ell}{d} + \frac{d}{2}\right).\end{aligned}$$

Then

$$m_c(p) = p \Pi_K(\ell_1, \ell_2) + (1 - p) \Pi_F(\ell_1, \ell_2),$$

$$D_c(p) = p(\alpha_1 - \beta_1) \Pi_K(\ell_1, \ell^*) - (1 - p) \beta_1 \Pi_F(\ell_1, \ell^*) \\ + p(\beta_2 - \alpha_2) \Pi_K(\ell^*, \ell_2) + (1 - p) \beta_2 \Pi_F(\ell^*, \ell_2),$$

with $\alpha_1 = s a + (1 - t) b - h_K \tau$, $\beta_1 = c_v + (1 - t) b$, $\alpha_2 = t b + (1 - s) a + h_K \tau$, $\beta_2 = t b - c_v$ (so that $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$, Theorem 1). As $d \rightarrow 0$, $m_c(p) \rightarrow \mathbb{1}[p \in S]$ for $p \notin \{q_1, q_2\}$ and $D_c(p) \rightarrow (V - \kappa)^+(p)$ for every p , the confounded values of Corollary 1.

Remark 2 (The Gaussian model is a calibration model). The hypotheses of Proposition 1 are a Gaussian order-flow model of the form of Paper 1's Definition D11 with any separation $d > 0$; they involve no value of d and no property of Paper 1's cells. Paper 1's ABIDES order-flow models were never shown to be Gaussian, so the Gaussian model is an *equivalent-Gaussian calibration model*, not Paper 1's cells, and Proposition 1 describes those cells only insofar as they are Gaussian with that separation. A separation d backed out of a Paper 1 accuracy through the Gaussian benchmark $\text{Acc}^* = \Phi(d/2)$ is a calibration heuristic only; wherever the closed forms of Proposition 1 are evaluated at such a d , d is also swept directly over a grid. Every statement about Paper 1's cells, including where targeted verification beats the best constant policy in them, rests on quantities computed on each cell's corrected classifier posteriors (Definition 5 applies to any order-flow model with finite densities), with no Gaussian assumption, and not on Proposition 1.

Proposition 1 shows how the mass and the saving depend on the separation d and on the prior through the logit shift ℓ , and it recovers the confounded values of Corollary 1 as $d \rightarrow 0$. Remark 2 fixes its scope: the Gaussian cell is a calibration model, and statements about the populations studied in Section 5 rest on the empirical quantities of Definition 5, not on Proposition 1.

4.9. Proofs

Proof of Lemma 1. Let $L_c(q) = qC_K + (1 - q)C_F$ (contain) and $L_\ell(q) = qH_K + (1 - q)H_F$ (leave). Their difference is affine:

$$D(q) := L_\ell(q) - L_c(q) = q(H_K - C_K) - (1 - q)(C_F - H_F) \\ = q a - (1 - q) b = (a + b)(q - q^*),$$

with $q^* = b/(a + b)$. By Assumption 1, $a, b > 0$, so $a + b > 0$ and $q^* \in (0, 1)$. Hence $D < 0$ (leave strictly cheaper) for $q < q^*$ and $D > 0$ (contain strictly cheaper) for $q > q^*$. Since $\min\{L_\ell, L_c\} = L_\ell - \max\{0, D\}$,

$$R(q) = L_\ell(q) - (a + b)(q - q^*)^+.$$

R is the minimum of two affine functions, hence continuous and concave; it is piecewise linear with slopes $H_K - H_F$ on $[0, q^*]$ and $C_K - C_F$ on $[q^*, 1]$, which differ by $-(a + b) \neq 0$, so there is exactly one kink, at q^* . $\square$

Proof of Lemma 2. Fix y with $P(Y = y) > 0$ (for densities, replace probabilities of Y by densities; the argument is unchanged), write $q = q(y)$, and let z be an outcome with $P(Z = z \mid Y = y) > 0$. By Bayes' rule and Assumption 2,

$$P(K \mid y, z) = \frac{P(z \mid K) P(y \mid K) P(K)}{\sum_{c \in \{K, F\}} P(z \mid c) P(y \mid c) P(c)}.$$

Divide numerator and denominator by $P(y) = \sum_c P(y \mid c) P(c)$ and use $P(y \mid c) P(c) / P(y) = P(c \mid y)$:

$$P(K \mid y, z) = \frac{P(z \mid K) q}{P(z \mid K) q + P(z \mid F) (1 - q)}.$$

For $z = +$ this is $qs / (qs + (1 - q)(1 - t)) = q^+$; for $z = -$ it is $q(1 - s) / (q(1 - s) + (1 - q)t) = q^-$. Likewise $P(Z = z \mid Y = y) = P(z \mid K) q + P(z \mid F) (1 - q)$, which is $P(+)$ or $P(-)$ of Definition 2 at $q = q(y)$. So, conditionally on $Y = y$, verification acts on the belief $q(y)$ exactly as in Definition 2 with the same (s, t) , and every quantity built from Definitions 2–3 (R_v , V , S) is evaluated at $q(y)$. Nothing in the argument uses the law of Y beyond $q(y)$; in particular it holds when $P(Y \mid K) = P(Y \mid F)$, in which case $q(y)$ equals the prior. Without Assumption 2 the factorisation in the first display fails and the update would depend on y through $P(z \mid y, c)$. $\square$

Proof of Lemma 3. Step 1 (homogeneous form). For an outcome z write $\alpha_z = P(z \mid K) q$ and $\beta_z = P(z \mid F) (1 - q)$, so $P(z) = \alpha_z + \beta_z$ and, by Definition 3, $P(z) R(q^z) = \min\{\alpha_z C_K + \beta_z C_F, \alpha_z H_K + \beta_z H_F\}$. The difference of the two arguments is $(\alpha_z H_K + \beta_z H_F) - (\alpha_z C_K + \beta_z C_F) = \alpha_z a - \beta_z b$, so

$$P(z) R(q^z) = (\alpha_z H_K + \beta_z H_F) - \max\{0, \alpha_z a - \beta_z b\}.$$

Here $\alpha_+ a - \beta_+ b = qsa - (1 - q)(1 - t)b = G^+(q)$ and $\alpha_- a - \beta_- b = q(1 - s)a - (1 - q)tb = -G^-(q)$. Summing over z and using $\alpha_+ + \alpha_- = q$, $\beta_+ + \beta_- = 1 - q$, with $L_\ell(q) = qH_K + (1 - q)H_F$:

$$P(+) R(q^+) + P(-) R(q^-) = L_\ell(q) - \max\{0, G^+(q)\} - \max\{0, -G^-(q)\}.$$

By the proof of Lemma 1, $R(q) = L_\ell(q) - \max\{0, D(q)\}$ with $D(q) = qa - (1-q)b$, and summing the two outcome terms gives $G^+(q) - G^-(q) = D(q)$. Therefore

$$V(q) = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}. \quad (1)$$

Step 2 (zeros and the value at q^).* G^+ is affine with slope $sa + (1-t)b > 0$ (as $a, b > 0$ and $s > 1-t \geq 0$); G^- is affine with slope $-(tb + (1-s)a) < 0$ (as $t > 1-s \geq 0$). Their unique zeros are q_L and q_U of the statement, and the formulas give $0 \leq q_L$ and $q_U \leq 1$. At q^* , $D(q^*) = 0$, so $G^+(q^*) = G^-(q^*) = q^*sa - (1-q^*)(1-t)b$. Since $q^*a = (1-q^*)b = ab/(a+b)$, this equals $(s+t-1)ab/(a+b)$, which is > 0 by Assumption 3. As G^+ is increasing with $G^+(q^*) > 0$, $q_L < q^*$; as G^- is decreasing with $G^-(q^*) > 0$, $q^* < q_U$. This proves (i).

Step 3 (the tent). Evaluate (1) on three ranges.

- $q \leq q_L$: $G^+ \leq 0$ and (as $q < q_U$) $G^- > 0$, so $G^+ - G^- < 0$; every term of (1) vanishes and $V = 0$.
- $q \geq q_U$: $G^- \leq 0$ and (as $q > q_L$) $G^+ > 0$, so $G^+ - G^- > 0$ and $V = G^+ - G^- - (G^+ - G^-) = 0$.
- $q_L < q < q_U$: $G^+ > 0$ and $G^- > 0$, so $V = G^+ - \max\{0, G^+ - G^-\} = \min\{G^+, G^-\} > 0$.

Hence $V = \max\{0, \min\{G^+, G^-\}\}$ on $[0, 1]$, and $V > 0$ exactly on (q_L, q_U) . On that interval, $G^+ < G^-$ iff $D < 0$ iff $q < q^*$. So $V = G^+$ (affine, strictly increasing) on $[q_L, q^*]$ and $V = G^-$ (affine, strictly decreasing) on $[q^*, q_U]$; V is continuous, and its unique maximiser is q^* , with $V(q^*) = (s+t-1)ab/(a+b) = (s+t-1)aq^*$. This proves (iii) and (iv).

Step 4 (posterior form of (ii)). For $q \in (0, 1)$, $P(+)>0$ and $P(-)>0$ because $s > 0$ and $t > 0$. Using $(a+b)q^* = b$,

$$(a+b)P(+)(q^+ - q^*) = (a+b)qs - bP(+) = qsa - (1-q)(1-t)b = G^+(q),$$

and in the same way $(a+b)P(-)(q^* - q^-) = G^-(q)$. So $G^+(q) > 0 \iff q^+ > q^*$ and $G^-(q) > 0 \iff q^- < q^*$; hence $q \in (q_L, q_U) \iff q^- < q^* < q^+$. At $q \in \{0, 1\}$, $V = 0$, and every defined posterior equals q , so the posteriors cannot straddle $q^* \in (0, 1)$: both sides of (ii) are false. $\square$

Proof of Theorem 1. Assumption 4 is what makes Definition 3 the correct verify-first risk: under K (probability q) verification adds $c_v + h_K\tau$ to the terminal cost, under F it adds c_v , the terminal costs and (s, t) are unchanged by the delay, and the terminal action is chosen optimally on the posterior, giving $c_v + qh_K\tau + \sum_z P(z)R(q^z)$. So $R_v < R$ iff $V > \kappa$, with $\kappa(q) = c_v + qh_K\tau$ affine and nondecreasing.

Step 1 ($S \subseteq (q_L, q_U)$). By Assumption 3, $\kappa(q) \geq 0$ on $[0, 1]$. By Lemma 3, $V = 0$ outside (q_L, q_U) , so $V > \kappa$ fails there.

Step 2 (S is an interval). Let $\varphi(q) = \min\{G^+(q), G^-(q)\} - \kappa(q)$ on $[0, 1]$. It is the minimum of two affine functions minus an affine function, hence concave and continuous. On (q_L, q_U) , $V = \min\{G^+, G^-\}$ (Lemma 3), and outside it $\min\{G^+, G^-\} \leq 0$ (since $G^+ \leq 0$ for $q \leq q_L$ and $G^- \leq 0$ for $q \geq q_U$), so $\varphi \leq -\kappa \leq 0$ there. Hence $S = \{q : \varphi(q) > 0\}$, a strict superlevel set of a continuous concave function: an open (relative to $[0, 1]$) interval, possibly empty. Since $\varphi(0) = \min\{-(1-t)b, tb\} - c_v \leq 0$ and $\varphi(1) = \min\{sa, -(1-s)a\} - c_v - h_K\tau \leq 0$, the endpoints are excluded and S is an open interval in $(0, 1)$.

Step 3 (non-emptiness criterion). On $[q^*, q_U]$, $\varphi = G^- - \kappa$ is strictly decreasing (G^- strictly decreasing, κ nondecreasing). On $[q_L, q^*]$, $\varphi = G^+ - \kappa$ is affine with slope $\sigma = sa + (1-t)b - h_K\tau$. If $\sigma \leq 0$, φ is nonincreasing on $[q_L, q_U]$, so $\sup_{(q_L, q_U)} \varphi \leq \varphi(q_L) = -\kappa(q_L) \leq 0$ and $S = \emptyset$. If $\sigma > 0$, φ is strictly increasing then strictly decreasing, so $\max_{[q_L, q_U]} \varphi = \varphi(q^*) = V(q^*) - \kappa(q^*)$. Either way, $S \neq \emptyset$ implies $\varphi(q^*) > 0$ (in the first case S is empty; in the second the maximum is at q^*). Conversely $\varphi(q^*) > 0$ gives $q^* \in S$. Hence $S \neq \emptyset \iff V(q^*) > c_v + q^*h_K\tau$, and then $q^* \in S$. Also, $S \neq \emptyset$ forces $\sigma > 0$: if $\sigma \leq 0$ we showed $S = \emptyset$.

Step 4 (thresholds). Assume $S \neq \emptyset$. Then $\sigma > 0$, and the left endpoint q_1 of S is the zero of the affine function $G^+ - \kappa = \sigma q - ((1-t)b + c_v)$ on $[q_L, q^*]$ if $\varphi(q_L) < 0$, or q_L itself if $\varphi(q_L) = 0$; in both cases $q_1 = (c_v + (1-t)b)/\sigma$ (when $\varphi(q_L) = 0$, $\kappa(q_L) = 0$, so $c_v = 0$ and $q_L h_K\tau = 0$, and the formula returns q_L). The right endpoint q_2 is the zero of $G^- - \kappa = tb - c_v - q(tb + (1-s)a + h_K\tau)$ on $(q^*, q_U]$, i.e. $q_2 = (tb - c_v)/(tb + (1-s)a + h_K\tau)$; its denominator is $\geq tb > 0$. As $G^+(q_1) = \kappa(q_1) \geq 0$ we get $q_1 \geq q_L$, and as $G^-(q_2) = \kappa(q_2) \geq 0$ we get $q_2 \leq q_U$; $q_1 < q^* < q_2$ because $q^* \in S$ and S is open.

Step 5 (policy). For $q \notin S$ the optimal action is the passive one of Lemma 1; since $q_1 < q^* < q_2$, this is leave on $q \leq q_1$ and contain on $q \geq q_2$. For $q \in S \subseteq (q_L, q_U)$, Lemma 3(ii) gives $q^- < q^* < q^+$, so by Lemma 1 the optimal terminal action after verification is contain on $Z = +$ and leave on

$Z = -$.

Step 6 ($c_v = 0, h_K \tau = 0$). Then $\kappa \equiv 0$ and $S = \{V > 0\} = (q_L, q_U)$ by Lemma 3(ii); the threshold formulas reduce to q_L and q_U . $\square$

Proof of Theorem 2. Throughout, only τ varies; V, q^*, q_L, q_U do not depend on τ (Assumption 4: s, t and the loss table are delay-free).

(i) *Nesting.* For $q \in [0, 1]$, $\kappa_\tau(q) = c_v + qh_K\tau$ is nondecreasing in τ because $qh_K \geq 0$. So $V(q) > \kappa_{\tau'}(q)$ implies $V(q) > \kappa_\tau(q)$ for $\tau < \tau'$, i.e. $S(\tau') \subseteq S(\tau)$. Now let $h_K > 0$ and $V(q^*) > c_v$. By (ii) below, whose proof does not use (i), $S(\tau) \neq \emptyset$ exactly for $\tau \in [0, \tau^*)$; so on $[0, \tau^*)$ the end points $q_1(\tau), q_2(\tau)$ are given by the formulas of Theorem 1 with positive denominators, and for $\tau \geq \tau^*$ the region is empty and Theorem 1's convention $q_1 = q_2 = q^*$ applies, so no monotonicity is claimed there. On $[0, \tau^*)$: $q_2 = (tb - c_v)/(tb + (1-s)a + h_K\tau)$ has a positive numerator (because $q_2 > q^* > 0$ and the denominator is positive) and a denominator strictly increasing in τ , so it is strictly decreasing. $q_1 = (c_v + (1-t)b)/(sa + (1-t)b - h_K\tau)$ has a positive denominator strictly decreasing in τ and a nonnegative numerator, zero iff $c_v = 0$ and $t = 1$; so q_1 is strictly increasing on $[0, \tau^*)$ unless $c_v = 0$ and $t = 1$, where $q_1 = 0$ on $[0, \tau^*)$.

(ii) *Critical delay.* By Theorem 1, $S(\tau) \neq \emptyset \iff V(q^*) > c_v + q^*h_K\tau$. With $h_K > 0$ and $q^* > 0$, this is $\tau < (V(q^*) - c_v)/(q^*h_K) = \tau^*$, which is > 0 when $V(q^*) > c_v$. By Lemma 3(iv), $V(q^*)/q^* = (s + t - 1)a$, giving the second expression for τ^* .

(iii) *Degenerate cases.* If $V(q^*) \leq c_v$, then $V(q^*) \leq c_v + q^*h_K\tau$ for every $\tau \geq 0$, so $S(\tau) = \emptyset$ by Theorem 1. If $h_K = 0$, κ does not involve τ , so neither does S .

(iv) *How the interval vanishes.* Let $h_K > 0, V(q^*) > c_v$, and $\tau \uparrow \tau^*$. The denominator of q_2 is continuous and positive at τ^* , so $q_2(\tau) \rightarrow q_2(\tau^*)$, the zero of $G^- - \kappa_{\tau^*}$. At τ^* , $G^-(q^*) = V(q^*) = \kappa_{\tau^*}(q^*)$, so that zero is q^* (it is unique: $G^- - \kappa_{\tau^*}$ is strictly decreasing). For q_1 : at τ^* the left-piece function $G^+ - \kappa_{\tau^*}$ vanishes at q^* and equals $-\kappa_{\tau^*}(q_L) = -(c_v + q_L h_K \tau^*)$ at q_L . If $c_v > 0$ or $t < 1$ (the latter giving $q_L > 0$), this is < 0 , so the slope $\sigma(\tau^*) = sa + (1-t)b - h_K\tau^*$ is > 0 , the formula for q_1 is continuous at τ^* , and $q_1(\tau) \rightarrow$ the zero of $G^+ - \kappa_{\tau^*}$, which is q^* . If $c_v = 0$ and $t = 1$, then $q_L = 0, G^+(q) = qsa$ and $\kappa_\tau(q) = qh_K\tau$, and $\tau^* = sa/h_K$ (as $s + t - 1 = s$). For $\tau < \tau^*$, $G^+ - \kappa_\tau = q(sa - h_K\tau) > 0$ on $(0, q^*]$, so $S(\tau) = (0, q_2(\tau))$ with $q_2(\tau) \downarrow q^*$ by (i) and the limit above; at τ^* , $G^+ - \kappa_{\tau^*} \equiv 0$ on $[0, q^*]$ and $S(\tau^*) = \emptyset$ by (ii). So the interval drops from $(0, q^*)$ (in the limit) to $\emptyset$. $\square$

Proof of Corollary 1. Reduction to a constant belief. By Assumption 2 the joint law of (M, Y, Z) is $P(M = m) P_m^Y(dy) P(Z = z | m)$. Let $\mu = P_K^Y = P_F^Y$. Then Y has law μ and is independent of (M, Z) , and $P(M = m, Z = z) = P(m) P(z | m)$. Write $P(z, K) = P(z | K) p$ and $P(z, F) = P(z | F) (1 - p)$ as in Definition 3. Conditioning on $Y = y$ in Definition 4 gives $J(\delta, \pi) = \int \ell(y) \mu(dy)$ with

$$\begin{aligned} \ell(y) = & (1 - \delta(y)) \left[\pi_0(y) (pC_K + (1 - p)C_F) \right. \\ & \left. + (1 - \pi_0(y)) (pH_K + (1 - p)H_F) \right] \\ & + \delta(y) \left[c_v + ph_K\tau + \sum_{z \in \{+, -\}} \left(\pi_z(y) (P(z, K)C_K + P(z, F)C_F) \right. \right. \\ & \left. \left. + (1 - \pi_z(y)) (P(z, K)H_K + P(z, F)H_F) \right) \right]. \end{aligned}$$

Here c_v and $h_K\tau$ are paid only on verification, the latter only under K (Assumption 4). Each expression of the form $\pi x + (1 - \pi) x'$ with $\pi \in [0, 1]$ is at least $\min\{x, x'\}$. So the first bracket is at least $R(p)$ (Definition 1). The second is at least $c_v + ph_K\tau + \sum_z \min\{P(z, K)C_K + P(z, F)C_F, P(z, K)H_K + P(z, F)H_F\} = R_v(p)$, by the reading of Definition 3. Hence, for every y ,

$$\ell(y) \geq (1 - \delta(y))R(p) + \delta(y)R_v(p) \geq \min\{R(p), R_v(p)\},$$

and $J \geq \min\{R(p), R_v(p)\}$. The constant policy that verifies iff $R_v(p) < R(p)$ and takes the minimizing terminal action in each case attains equality. So

$$R_{Y,v}^* = \min\{R(p), R_v(p)\}, \quad R_Y^* = R(p)$$

(the second with $\delta \equiv 0$). The attaining policy depends on p only, not on y . When $p \in S$ it is the verify branch of Theorem 1 at $q = p$: contain on $Z = +$ and leave on $Z = -$, because $p^- < q^* < p^+$ by Lemma 3(ii). Otherwise it is the passive rule of Lemma 1.

(i) By Assumption 1, $\min\{C_K, H_K\} = C_K$ and $\min\{C_F, H_F\} = H_F$, so $R_{\text{oracle}} = pC_K + (1 - p)H_F$. Subtracting it from both terms of $R(p)$ gives $\text{CoA}^* = \min\{(1 - p)(C_F - H_F), p(H_K - C_K)\} = \min\{pa, (1 - p)b\}$. This is > 0 since $a, b > 0$ and $0 < p < 1$.

(ii) By Definition 3, $R_v(p) = \kappa(p) + R(p) - V(p)$. So

$$\text{CoA}_v^* = \min\{R(p), R_v(p)\} - R_{\text{oracle}} = \text{CoA}^* - (V(p) - \kappa(p))^+.$$

With $G^\pm$ as in Lemma 3,

$$\begin{aligned}\text{CoA}^* - E(p) &= \min\{pa - p(1-s)a - (1-p)(1-t)b, \\ &\quad (1-p)b - p(1-s)a - (1-p)(1-t)b\} - \kappa(p) \\ &= \min\{G^+(p), G^-(p)\} - \kappa(p).\end{aligned}$$

By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$, and $\kappa(p) \geq 0$ by Assumption 3. Suppose $\min\{G^+, G^-\} > \kappa$. Then $V = \min\{G^+, G^-\}$ and $(V - \kappa)^+ = \min\{G^+, G^-\} - \kappa$. Otherwise $V \leq \max\{0, \kappa\} = \kappa$, and $(V - \kappa)^+ = 0 = (\min\{G^+, G^-\} - \kappa)^+$. In both cases $(V - \kappa)^+ = (\text{CoA}^* - E)^+$, so $\text{CoA}_v^* = \text{CoA}^* - (\text{CoA}^* - E)^+ = \min\{\text{CoA}^*, E(p)\}$. (Direct check: $E(p)$ is the cost of the verify branch above minus R_{oracle} . That is c_v , plus $h_K\tau$ under K , a miss a with probability $1-s$ under K , and a false containment b with probability $1-t$ under F .)

(iii) By (ii), $\text{CoA}_v^* < \text{CoA}^* \iff V(p) > \kappa(p) \iff p \in S$ (Definition 3). By Theorem 1, $S \neq \emptyset \iff V(q^*) > c_v + q^*h_K\tau$, and then $S = (q_1, q_2)$. If $h_K > 0$ and $V(q^*) > c_v$, Theorem 2(ii) turns the non-emptiness condition into $\tau < \tau^*$. If $V(q^*) \leq c_v$, then $S(\tau) = \emptyset$ for all $\tau \geq 0$ by Theorem 2(iii).

(iv) The reduction is $D(p) = (V(p) - \kappa(p))^+$, and $V - \kappa \leq -\kappa \leq 0$ outside (q_L, q_U) . On $[q^*, q_U]$, $V - \kappa = G^- - \kappa$ is strictly decreasing (Lemma 3(iii); κ is nondecreasing). On $[q_L, q^*]$, $V - \kappa$ is affine. It is ≤ 0 at q_L and equals $V(q^*) - \kappa(q^*)$ at q^* . So its maximum over $[q_L, q^*]$ is attained at an endpoint, and $\max_p D(p) = (V(q^*) - \kappa(q^*))^+ = (V(q^*) - c_v - q^*h_K\tau)^+$, attained at q^* . When this maximum is positive, D is strictly increasing on $[q_L, q^*]$ (an affine function going from ≤ 0 to a positive value) wherever it is positive, and strictly decreasing on $[q^*, q_U]$. So the maximizer is unique.

(v) Since $\text{CoA}^* > 0$ by (i), $\text{CoA}_v^* = 0$ iff $E(p) = 0$. Each of the four terms of $E(p)$ is nonnegative (Assumption 3). With $p \in (0, 1)$ and $a, b > 0$, all four vanish iff $c_v = 0$, $h_K\tau = 0$, $s = 1$ and $t = 1$. $\square$

Proof of Remark 1. Step 1 of the proof of Lemma 3 used only Definitions 1–3 and holds without Assumptions 1 and 3: $V = \max\{0, G^+\} + \max\{0, -G^-\} - \max\{0, G^+ - G^-\}$, with $G^+ - G^- = D = qa - (1-q)b$.

(a) If $s + t = 1$, then $P(z | K) = P(z | F)$ for both outcomes, so $q^z = q$ whenever $P(z) > 0$. Then $\sum_z P(z)R(q^z) = R(q)$ (the $P(z) = 0$ term is zero by Definition 3), so $V \equiv 0$. As $\kappa \geq 0$, $V > \kappa$ never holds and $S = \emptyset$.

(b1) If $a \leq 0 \leq b$, then $D(q) = qa - (1-q)b \leq 0$ on $[0, 1]$, so leaving is weakly optimal everywhere and $R = L_\ell$ is affine. Each outcome term satisfies

$\alpha_z a - \beta_z b \leq 0$ (as $\alpha_z, \beta_z \geq 0$), so $\sum_z P(z)R(q^z) = \sum_z (\alpha_z H_K + \beta_z H_F) = L_\ell(q) = R(q)$ and $V \equiv 0$. The case $b \leq 0 \leq a$ is symmetric ($D \geq 0$, contain weakly optimal everywhere, each $\alpha_z a - \beta_z b \geq 0$, so $\sum_z P(z)R(q^z) = L_c(q) = R(q)$). In both, $S = \emptyset$ since $\kappa \geq 0$. These two sign patterns cover every failure of Assumption 1 except $a < 0$ and $b < 0$: if one of a, b is ≤ 0 and the other is ≥ 0 , one of the two patterns holds, and the only remaining way to have $a \leq 0$ or $b \leq 0$ is $a < 0$ and $b < 0$.

(b2) Let $a < 0$ and $b < 0$. Then $a + b < 0$ and $q^* = b/(a + b) \in (0, 1)$, and $D(q) = (a + b)(q - q^*)$ is > 0 (contain strictly cheaper) for $q < q^*$ and < 0 for $q > q^*$. Exchange the names of the two terminal actions: the new costs are $C'_K = H_K$, $C'_F = H_F$, $H'_K = C_K$, $H'_F = C_F$, so $a' = H'_K - C'_K = -a > 0$, $b' = C'_F - H'_F = -b > 0$ and $b'/(a' + b') = q^*$: the renamed problem satisfies Assumption 1. R is the minimum over the two actions (Definition 1) and each term $P(z)R(q^z)$ is the minimum of the two joint costs (Definition 3), so R , V and S do not depend on the names; κ involves no terminal action; Assumptions 2–4 do not refer to the names (Assumption 4 charges $h_K \tau$ whichever terminal action is taken). In Definition 4, renaming maps a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$ to $(\delta, 1 - \pi_0, 1 - \pi_+, 1 - \pi_-)$, a bijection that preserves J , and R_{oracle} is symmetric in the two actions. Hence every result stated under Assumption 1 holds for the renamed problem with $(a', b') = (-a, -b)$; read back in the original names, contain and leave are exchanged in each policy, which gives the rule stated in (b2).

(c) Take $c_v < 0$, $t < 1$ (so $q_L > 0$ under Assumption 1), $h_K > 0$, and $\tau > 0$ small enough that $V(q^*) > q^* h_K \tau$. Choose $c_v \in (-q_L h_K \tau, 0)$. Then $q^* \in S$, because $V(q^*) > q^* h_K \tau > \kappa(q^*)$. At $q = 0$, $V(0) = 0 > c_v = \kappa(0)$, so $0 \in S$. At $q = q_L$, $V(q_L) = 0 < c_v + q_L h_K \tau = \kappa(q_L)$, so $q_L \notin S$. Since $0 < q_L < q^*$, S is not an interval. $\square$

Proof of Lemma 4. Write $f_K = f_K^c$, $f_F = f_F^c$, $f_p = p f_K + (1 - p) f_F$ (the ν -density of P_p), $q = q_p$, and $g = (V - \kappa)^+$. Let $N = \{y : f_p(y) = 0\}$; then $P_p(N) = 0$, and q is a measurable map from N^c to $[0, 1]$.

Properties of g . By Lemma 3, $V = \max\{0, \min\{G^+, G^-\}\}$ with $G^\pm$ affine, so V is continuous on $[0, 1]$ and $0 \leq V \leq V(q^*)$; $\kappa(q) = c_v + q h_K \tau$ is affine and ≥ 0 (Assumption 3). Hence g is continuous, $0 \leq g \leq V \leq V(q^*)$, and $g > 0$ exactly on S (Definition 3). At $q = 0$: $G^+(0) = -(1 - t)b \leq 0$, so $V(0) = 0$ and $g(0) = 0$.

(i) *Saving.* Fix a verify-first policy $(\delta, \pi_0, \pi_+, \pi_-)$. By Assumption 2 the joint law of (M, Y, Z) has density $P(M = m) f_m(y) P(Z = z | m)$ with

respect to (counting $\times \nu \times$ counting). Integrating out m and z in Definition 4,

$$J(\delta, \pi) = \int_{N^c} f_p(y) \ell_{q(y)}(y) \nu(dy),$$

where $\ell_q(y)$ is the expression $\ell(y)$ of the proof of Corollary 1 with p replaced by q , since $pf_K(y)/f_p(y) = q(y)$ and $(1-p)f_F(y)/f_p(y) = 1-q(y)$. The pointwise argument of that proof gives $\ell_{q(y)}(y) \geq (1-\delta(y))R(q(y)) + \delta(y)R_v(q(y)) \geq \min\{R, R_v\}(q(y))$, and, for passive policies ($\delta \equiv 0$), $\ell_{q(y)}(y) \geq R(q(y))$. All costs are bounded, so every integral is finite. Equality holds for the following measurable policy: $\delta(y) = \mathbb{1}[q(y) \in S]$ (S is an interval, Theorem 1); $\pi_0(y) = \mathbb{1}[q(y) > q^*]$ (Lemma 1); $\pi_+ \equiv 1$, $\pi_- \equiv 0$. The last two are optimal on $\{q(y) \in S\}$: there $V(q) > \kappa(q) \geq 0$, so $G^+(q) > 0$ and $G^-(q) > 0$ (Lemma 3). Since $G^+(q) = P(+, K)a - P(+, F)b$, containing is strictly better than leaving after $Z = +$. Since $G^-(q) = P(-, F)b - P(-, K)a$, leaving is strictly better after $Z = -$. With $\delta \equiv 0$ and the same π_0 , equality holds in the passive bound. Hence

$$R_{Y,v}^* = \mathbb{E}_p[\min\{R, R_v\}(q(Y))], \quad R_Y^* = \mathbb{E}_p[R(q(Y))].$$

By Definition 3, $R - R_v = V - \kappa$, so $R - \min\{R, R_v\} = (V - \kappa)^+ = g$, and $R_Y^* - R_{Y,v}^* = \mathbb{E}_p[g(q(Y))] = D_c(p)$. Subtracting R_{oracle} from both infima gives $D_c(p) = \text{CoA}_c^*(p) - \text{CoA}_{v,c}^*(p)$. For the bounds: $D_c \geq 0$ since $g \geq 0$. $\text{CoA}_{v,c}^* \geq 0$ because, for every policy, the integrand of J is at least $\min\{L(\text{contain}, M), L(\text{leave}, M)\}$ ($c_v \geq 0$, $h_K\tau \geq 0$), and the expectation of that is R_{oracle} ; so $D_c \leq \text{CoA}_c^*$. Finally $g \leq V(q^*)\mathbb{1}_S$ gives $D_c(p) \leq V(q^*)m_c(p)$.

Pointwise limit under F . Let y satisfy $f_F(y) > 0$ (a set of full P_F^c -measure). Then $y \notin N$ for every $p \in (0, 1)$, and

$$0 \leq q_p(y) \leq \frac{p f_K(y)}{(1-p) f_F(y)} \longrightarrow 0 \quad (p \rightarrow 0),$$

because $f_K(y) < \infty$.

(ii) Split $D_c(p) = p \mathbb{E}_K[g(q_p(Y))] + (1-p) \mathbb{E}_F[g(q_p(Y))]$, where $\mathbb{E}_m$ is the expectation under P_m^c (the event N is null under both). The first term is at most $p V(q^*) \rightarrow 0$. In the second, $g(q_p(y)) \rightarrow g(0) = 0$ for P_F^c -almost every y , by the pointwise limit and continuity of g . The integrand is bounded by the constant $V(q^*)$, so dominated convergence gives $\mathbb{E}_F[g(q_p(Y))] \rightarrow 0$. Hence $D_c(p) \rightarrow 0$.

(iii) If $S = \emptyset$, then $m_c \equiv 0$. Let $S = (q_1, q_2) \neq \emptyset$. By Theorem 1 the denominator of q_1 is positive, so $q_1 > 0$ iff $c_v + (1 - t)b > 0$, i.e. iff $c_v > 0$ or $t < 1$ ($b > 0$, Assumption 1). Suppose $q_1 > 0$. Then $m_c(p) \leq p + (1 - p)P_F^c(q_p(Y) > q_1)$. For P_F^c -almost every y , $q_p(y) \rightarrow 0 < q_1$, so $\mathbb{K}[q_p(y) > q_1] \rightarrow 0$. Dominated convergence (bound 1) gives $P_F^c(q_p(Y) > q_1) \rightarrow 0$, hence $m_c(p) \rightarrow 0$.

(iv) Let $S \neq \emptyset$, $c_v = 0$ and $t = 1$. Then $q_1 = 0$ (Theorem 1) and $S = (0, q_2)$ with $q_2 > q^* > 0$. Write $m_c(p) = pP_K^c(q_p(Y) \in S) + (1 - p)P_F^c(q_p(Y) \in S)$; the first term is at most $p \rightarrow 0$. Take y with $f_F(y) > 0$. If $f_K(y) = 0$, then $q_p(y) = 0 \notin S$ for every p . If $f_K(y) > 0$, then $q_p(y) > 0$, and $q_p(y) < q_2$ for all small enough p by the pointwise limit, so $q_p(y) \in S$ eventually. Thus $\mathbb{K}[q_p(y) \in S] \rightarrow \mathbb{K}[f_K(y) > 0]$ for P_F^c -almost every y . Dominated convergence gives $P_F^c(q_p(Y) \in S) \rightarrow P_F^c(f_K(Y) > 0)$, and $(1 - p) \rightarrow 1$. $\square$

Proof of Proposition 1. Law of the log-likelihood ratio. Let $\Delta = \mu_K - \mu_F$ and $\bar{\mu} = (\mu_K + \mu_F)/2$. The Gaussian densities are positive everywhere, and

$$\Lambda(y) = \log \frac{f_K(y)}{f_F(y)} = \Delta^\top \Sigma_Y^{-1}(y - \bar{\mu}).$$

Under $M = K$, $y - \bar{\mu} \sim \mathcal{N}(\Delta/2, \Sigma_Y)$, so $\Lambda \sim \mathcal{N}(d^2/2, d^2)$. Under $M = F$, $\Lambda \sim \mathcal{N}(-d^2/2, d^2)$. By Definition 5, $\text{logit } q_p(Y) = \ell + \Lambda(Y)$, which is strictly increasing in Λ . For $-\infty \leq u < v \leq \infty$, with Λ continuous ($d > 0$),

$$P_K^c(\text{logit } q_p(Y) \in (u, v)) = P\left(\frac{d^2}{2} + dW \in (u - \ell, v - \ell)\right) = \Pi_K(u, v),$$

with $W \sim \mathcal{N}(0, 1)$, and likewise $P_F^c(\cdot) = \Pi_F(u, v)$. Open and half-open intervals have the same probability.

Mass. $q_p(Y) \in S = (q_1, q_2)$ iff $\text{logit } q_p(Y) \in (\ell_1, \ell_2)$ (with the stated conventions when $q_1 = 0$ or $q_2 = 1$; $q_p(Y) \in (0, 1)$ always). Since $P_p = pP_K^c + (1 - p)P_F^c$, this gives m_c .

Saving. On S , $V > \kappa \geq 0$, so $V = \min\{G^+, G^-\}$ (Lemma 3). $G^+ - G^- = qa - (1 - q)b$ is ≤ 0 iff $q \leq q^*$. So on $(q_1, q^*]$, $g := (V - \kappa)^+ = G^+ - \kappa = \alpha_1 q - \beta_1$; on $[q^*, q_2)$, $g = G^- - \kappa = \beta_2 - \alpha_2 q$. Outside S , $g = 0$. Expanding $G^\pm$ from Lemma 3 and $\kappa = c_v + qh_K\tau$ gives the stated α_i, β_i . Theorem 1 gives $q_1 = \beta_1/\alpha_1$ and $q_2 = \beta_2/\alpha_2$. For any Borel $A \subseteq (0, 1)$,

$$\mathbb{E}_p[q_p(Y) \mathbb{K}_A(q_p(Y))] = \int \frac{pf_K}{f_p} \mathbb{K}_A(q_p) f_p d\nu = pP_K^c(q_p(Y) \in A).$$

Take $A_1 = (q_1, q^*]$ and $A_2 = (q^*, q_2)$, and write $P_p = pP_K^c + (1-p)P_F^c$. Then

$$\mathbb{E}_p[(\alpha_1 q_p - \beta_1) \mathbb{K}_{A_1}(q_p)] = p(\alpha_1 - \beta_1) P_K^c(A_1) - (1-p) \beta_1 P_F^c(A_1),$$

$$\mathbb{E}_p[(\beta_2 - \alpha_2 q_p) \mathbb{K}_{A_2}(q_p)] = p(\beta_2 - \alpha_2) P_K^c(A_2) + (1-p) \beta_2 P_F^c(A_2),$$

where $P_m^c(A)$ abbreviates $P_m^c(q_p(Y) \in A)$. Substituting $P_m^c(A_1) = \Pi_m(\ell_1, \ell^*)$ and $P_m^c(A_2) = \Pi_m(\ell^*, \ell_2)$, and summing, gives D_c .

Confounded limit $d \rightarrow 0$. For fixed finite $u < v$ and $\ell \notin \{u, v\}$, $(v - \ell)/d \mp d/2 \rightarrow \pm\infty$ according to the sign of $v - \ell$. So $\Pi_K(u, v), \Pi_F(u, v) \rightarrow \mathbb{K}[u < \ell < v]$, and the same holds with an infinite endpoint. At $\ell = u$ or $\ell = v$ the corresponding Φ term tends to $\Phi(0) = 1/2$. Hence $m_c(p) \rightarrow \mathbb{K}[\ell_1 < \ell < \ell_2] = \mathbb{K}[p \in S]$ for $p \notin \{q_1, q_2\}$. For D_c , take $p \in (q_1, q^*)$: the limit is $p(\alpha_1 - \beta_1) - (1-p)\beta_1 = \alpha_1 p - \beta_1 = g(p)$. Take $p \in (q^*, q_2)$: the limit is $\beta_2 - \alpha_2 p = g(p)$. Take $p \notin [q_1, q_2]$: the limit is $0 = g(p)$. At $p = q^*$ each of the two pieces tends to one half of its bracket. The brackets are $\alpha_1 q^* - \beta_1$ and $\beta_2 - \alpha_2 q^*$, and both equal $g(q^*)$ by continuity of $V - \kappa$. So the limit is $g(q^*)$. At $p = q_1 > 0$ the limit is $\frac{1}{2}(\alpha_1 q_1 - \beta_1) = 0 = g(q_1)$. At $p = q_2 < 1$ it is $\frac{1}{2}(\beta_2 - \alpha_2 q_2) = 0 = g(q_2)$. These are the confounded values $\mathbb{K}[p \in S]$ and $(V - \kappa)^+(p)$ of Corollary 1(ii)–(iii). $\square$

5. Empirical study on Paper 1's cells

This section is exploratory. Its design was fixed after the cost table of Paper 1 [1] for the same cells had been examined, so its outcome is reported as exploratory evidence, not as a pre-registered test.

5.1. Data and passive posteriors

[To be written from the lab's manifest.] Passive posteriors are the order-flow classifier probabilities of Paper 1, refitted exactly as Paper 1 fits them, on each design and in both the summary-matched and the contrast regime, and moved to each cell's prior by the prior-shift correction [18]. A cell is a design, a regime and a prior. The failure cells are those in which Paper 1 reports that its classifier does not beat the best constant policy; they are read from Paper 1's results, not re-derived here.

[Pending brief paper2/B001: table of cells, priors and failure-cell flags.]

5.2. Verification parameters and plausible set

[To be written from the lab's manifest.] Verification outcomes enter as an exact expectation over the outcome Z given the true cause. The grid covers sensitivity and specificity, the fixed cost of the check expressed as a share of $a q^*$, the peak value of information of a perfect check (Lemma 3(iv)), and the harm accrued during the delay expressed as a share of a ; in these units the condition of Theorem 1 compares their sum with the Youden index $s + t - 1$. A corner of near-free checks that almost never flag a fault is set aside as implausible before any result, because it drives q_1 to zero (Lemma 4(iv)).

[Pending brief paper2/B001: grid values and plausible set.]

5.3. Policies compared

Five policies are compared on the same cells: always contain, always leave, the passive classifier thresholded at q^* (Lemma 1), verify everything, and targeted verification by the rule of Theorem 1. For every policy the false-positive and false-negative rates are reported beside the expected cost, because the prior-shift correction trades one for the other.

5.4. Verification interval and critical delay

[Pending.] Per cell and grid point: q_1 , q_2 and τ^* in closed form, cross-checked numerically against the rule; the location of the corrected posteriors relative to the interval.

[Pending brief paper2/B001: verification-interval map and critical delay.]

5.5. Targeted verification in the failure cells

[Pending.] Primary exploratory analysis. Per failure cell and grid point: expected cost of targeted verification minus that of the best constant policy, with bootstrap confidence intervals over test instances; the region predicted from Theorem 1, where q_1 lies below the upper quantiles of the cell's corrected posteriors; and the agreement between the empirical and the predicted regions. Outcome recorded as supports, weakens, refutes or inconclusive at this power, as stated in the brief.

[Pending brief paper2/B001: failure-cell region, confidence intervals and agreement.]

5.6. Mass, saving and recovered share on all cells

[Pending.] Per cell: the verification mass m_c , the saving D_c beside its plug-in value, and the recovered share ρ_c of Definition 5, with confidence intervals. Descriptive; decides no claim.

[Pending brief paper2/B001: per-cell mass, saving and recovered share.]

5.7. Policy comparison

[Pending.] Expected cost, FPR and FNR of the five policies on all cells.

[Pending brief paper2/B001: policy comparison with FPR/FNR.]

5.8. Sensitivity analyses

[Pending.] Passive threshold computed with the measured leave-a-fault cost instead of $H_F = 0$; optional Gaussian cross-check of Proposition 1, labelled as a calibration model (Remark 2), with the separation also swept directly.

[Pending brief paper2/B001: sensitivity analyses.]

6. Discussion

6.1. What the empirical study shows

[Pending brief paper2/B001; to be written from the recorded outcome, as exploratory evidence.]

6.2. When the separate channel fails

Every result of Section 4 that involves order flow rests on Assumption 2. If the outcome of the check depends on the order flow given the cause, the verification update is no longer a function of the passive posterior alone and Lemma 2 does not hold. The assumption is a modelling choice about what the check reads, and its fit to a particular verification procedure has to be argued for that procedure.

6.3. One check, then act

The rule is derived for a single check followed by a terminal action (Assumption 3), with the delay charged in full on top of the terminal cost and no order flow used while the check runs (Assumption 4). Repeated checks, graded responses and evidence that keeps arriving during the delay turn the problem into a sequential one, which this paper does not treat.

6.4. Posteriors and calibration

The saving identity of Lemma 4 holds for the true posterior. Applied to classifier scores it holds only as far as the corrected scores are calibrated, which is why the empirical study reports the plug-in saving beside the realised one.

7. Conclusion

Verification that reads authorization is not bound by what order flow can reveal, and it composes with any passive posterior. Transposing the threshold approach, a check pays exactly on an interval of beliefs around the passive containment threshold, and because a true compromise keeps doing harm while the check runs, that interval vanishes at a critical delay in closed form. The interval does not depend on the prior; the saving it brings to a population of cases does, and vanishes as compromise becomes rare.

[Pending brief paper2/B001: one paragraph on the exploratory result in Paper 1's failure cells.]

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